

PPL Meta Mini Age Estimation Backend - User Manual

Overview

The **PPL Meta Mini Age Estimation Backend** is a headless application designed to analyze short video clips (1-3 seconds) and provide age estimation results to determine whether detected individuals are adults or underaged persons.

Application Scope

- **Input:** Short video files (1-3 seconds duration recommended)
 - **Output:** Age estimation analysis with adult/underaged classification
 - **Deployment:** Dockerized headless service with REST API
 - **Platform:** Linux containers (compatible with Windows Docker Desktop VM)
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Installation & Setup

Prerequisites

- Docker Desktop installed and running
- Minimum 2GB RAM available for the container
- Port 8004 (or custom port) available

Download & Installation

1. Download the application:

- Download from: <https://we.tl/t-5SGZZtgCrl>
- Available until: Fri 31 Jul 2025 (link will be updated)
- File: `ppl-meta-mini-beta085.tar` (1.5GB)

2. Load the Docker image:

Linux/macOS:

```
docker load -i ppl-meta-mini-beta085.tar
```

Windows Command Prompt:

```
docker load < ppl-meta-mini-beta085.tar
```

3. Run the application:

Command Line:

```
docker run -d --name ppl-meta-mini -p 8004:8004 nickglezakos/ppl-meta-mini-beta085:latest
```

Docker Desktop GUI:

- Navigate to Images → `nickglezakos/ppl-meta-mini-beta085:latest`
- Click “Run”
- Set Host Port: 8004 (or your preferred port like 8005)
- Click “Run”

4. Access the application:

- Swagger UI: <http://localhost:8004/docs>
 - API Base URL: <http://localhost:8004/api/v1/>
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API Usage

Primary Endpoint

POST /api/v1/upload-and-analyze

Example Request

```
curl -X 'POST' \
  'http://localhost:8004/api/v1/upload-and-analyze?max_faces_per_frame=10&proximity_threshold=50&confidence_threshold=0.5' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@your-video.mp4;type=video/mp4'
```

Parameters

Parameter	Type	Default	Description
max_faces_per_frame	int	10	Maximum faces to detect per frame. Recommended: Keep at 10
proximity_threshold	float	50	Placeholder parameter (no effect on processing)
confidence_threshold	float	0.5	Face detection confidence. Recommended: 0.5
frame_interval	int	15	Frame sampling interval. Recommended: 3-5 for 1-2 second videos
file	file	-	Video file (MP4 recommended, 1-2 seconds duration)

Parameter Guidelines

- **confidence_threshold:**
 - < 0.5: May produce false positives
 - 0.5: Recommended balance
 - > 0.6: May miss faces in lower quality frames
- **frame_interval:**
 - For 1-2 second videos: Use 3-5
 - Higher values = faster processing, fewer samples
 - Lower values = more thorough analysis, slower processing

Response Format

Successful Response Example

```
{
  "persons": {
    "1": {
      "face_id": "frame_120_face_1",
      "frame_number": 120,
      "quality_score": 0.83,
      "bbox": [88, 410, 413, 735],
      "age_detection": {
```

```

        "estimated_age": 37
    },
    "distance": 9.47
},
"2": {
    "face_id": "frame_165_face_1",
    "frame_number": 165,
    "quality_score": 0.786,
    "bbox": [44, 409, 388, 753],
    "age_detection": {
        "estimated_age": 36
    },
    "distance": 8.45
}
},
"file_info": {
    "storage_path": "/tmp/ppl_mini_video_processing/preprocessed_1753891163_video.mp4"
}
}

```

Response Fields

Field	Description
face_id	Unique identifier for detected face
frame_number	Frame where face was detected
quality_score	Face quality assessment (0.0-1.0)
bbox	Bounding box coordinates [x1, y1, x2, y2]
estimated_age	Predicted age in years
distance	Relative distance from camera (analogous units)
storage_path	Temporary file location

Age Classification Logic

Underaged Detection Criteria

Trigger an underaged alert when **ALL** conditions are met:

1. estimated_age < 30
2. distance <= 10 AND distance > 2
3. quality_score > 0.250

Important Notes

- **Distance < 2:** Likely false positive (too close/distorted)
- **Distance > 10:** Low confidence due to distance
- **Quality Score < 0.250:** Insufficient image quality for reliable estimation

Age Estimation Accuracy

- **Optimal conditions:** Near camera, good lighting, no motion blur
- **Real-world deviation:** Age estimates deviate from actual age based on:
 - Distance from camera
 - Lighting conditions

- Motion blur
 - Frame quality
 - Facial angle and expression
 - Women (heavy) makeup
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Best Practices

Video Recommendations

- **Duration:** 1-2 seconds optimal
- **Sampling (frame interval):** frame interval 3 or close. Use > 3 only for very “weak” machines
- **Quality:** Good lighting, minimal motion blur
- **Format:** MP4 recommended
- **Resolution:** Standard definition sufficient
- **Content:** Clear facial visibility

Performance Optimization

- Use recommended parameter values
- Shorter videos process faster
- Good lighting improves accuracy
- Stable camera reduces false detections

Integration Guidelines

- **Primary person:** Use the person with the smallest **distance** value
 - **Multiple detections:** Same person may appear multiple times; use closest detection
 - **Confidence threshold:** Combine **quality_score**, **distance**, and **estimated_age** for decisions
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Data Management

Automatic Cleanup

- Application automatically stores processed videos at **storage_path**
- **Auto-deletion:** Only the 3 most recent videos are retained
- **Cleanup trigger:** Runs on each new video upload

Quality Assurance & Training

For algorithm improvement and feedback:

1. **Retain:** Last 3 processed videos and their JSON responses
 2. **Provide feedback:** Submit videos with actual age verification
 3. **Include:** Both **storage_path** files and corresponding API responses
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Troubleshooting

Common Issues

Port already in use:

```
# Use different port
docker run -d --name ppl-meta-mini -p 8005:8004 nickglezakos/ppl-meta-mini-beta085:latest
# Access at http://localhost:8005/docs
```

Container won't start:

```
# Check logs
docker logs ppl-meta-mini
```

```
# Restart container
docker restart ppl-meta-mini
```

No faces detected: - Check video quality and lighting - Reduce `confidence_threshold` to 0.4 - Ensure faces are clearly visible - Try shorter `frame_interval` (e.g., 2)

Health Check

Test the service is running:

```
curl http://localhost:8004/health
```

Expected response:

```
{"status": "healthy", "service": "ppl-meta-mini"}
```

Technical Specifications

- **Container Size:** ~1.5GB
 - **Memory Usage:** ~512MB-1GB during processing
 - **Platform:** Linux AMD64 (Windows Docker Desktop compatible)
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Support & Updates

- **Current Version:** Beta085
 - **Download Link:** <https://we.tl/t-5SGZZtgCr1> (expires Fri 31 Jul 2025)
 - **API Documentation:** Available at `/docs` endpoint when running
 - **Updates:** Link will be refreshed before expiration
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For technical support or algorithm training data submission, please provide the processed video files and their corresponding JSON responses.